

MPA vs SPA (Multi Page Application vs Single Page Application)

Web applications are mainly divided into two types:

1. MPA - Multi Page Application
2. SPA - Single Page Application

Both are used to build websites and web applications but work differently.

1. Multi Page Application (MPA)

In MPA, every user request loads a completely new HTML page from the server. Whenever the user clicks on a link, the browser refreshes and loads another page.

Characteristics of MPA:

- Full page reload happens
- Each page has separate HTML
- Traditional website structure
- Easier SEO optimization
- Slower navigation compared to SPA

Examples of MPA:

- Amazon
- Flipkart
- Traditional banking websites
- College portals

2. Single Page Application (SPA)

In SPA, only one HTML page is loaded initially. After that, content updates dynamically without refreshing the full page. JavaScript frameworks mainly power SPA applications.

Characteristics of SPA:

- No full page reload
- Faster navigation
- Better user experience
- Uses APIs for data fetching
- Heavy use of JavaScript

Examples of SPA:

- Gmail
- Facebook
- Instagram
- Netflix
- React Applications

3. Difference Between MPA and SPA

Feature	MPA	SPA
Page Loading	Reloads full page	Updates only content
Speed	Slower	Faster
User Experience	Traditional	Smooth and modern
SEO	Better	Can be difficult
Frontend Framework	Optional	Mostly required
Server Requests	Many page requests	Mostly API requests
Examples	Amazon, Flipkart	Facebook, Gmail

4. When to Use MPA and SPA

Use MPA when:

- Website has many independent pages
- SEO is very important
- Traditional websites are needed

Use SPA when:

- Fast user interaction is needed
- Real-time updates are important
- Building dashboards or modern web apps

5. Technologies Used

MPA Technologies:

- PHP
- JSP
- ASP.NET

SPA Technologies:

- React
- Angular
- Vue

6. Final Summary

MPA loads a new page every time the user navigates. SPA loads one page and updates content dynamically. Modern applications mostly prefer SPA because of better speed and user experience.